corrosion can be defined as any process of deterioration (or destruction) and consequent loss of solid metallic material through an unwanted chemical or electro chemical attack by its environment, starting at its surface

example

① Rusting of iron

rusting of iron ~~ohe~~ is formed when exposed to the atmospheric condition

A layer of reddish scale or power of oxide ($FeSO_4$) is formed - $Fe_3O_4$

② corrosion of cu

formation of green film of basic carbonate [$CuCO_3 + Cu(OH)_2$] on the surface of copper, when exposed to moist air containing carbon dioxide

# Types of corrosion

1. Dry corrosion
2. Wet corrosion

## 1. Dry or chemical or Atmospheric corrosion

- Dry corrosion occurs mainly by direct attack of atmospheric gases such as oxygen, halogens, hydrogen sulphide, nitrogen, and sulphur dioxide. etc. on the surface of metals.
- The metal forms corresponding compounds such as oxides, sulphides, carbonates etc.
- This type of corrosion forms solid layer of corrosion products of different kinds
- If the layer formed is very thin it is called 'film' and if it exceeds this value it is called 'scale'.
- The action of some organic or anhydrous inorganic liquid metals also cause this type of corrosion.

This type of corrosion is divided into following classes.

i] corrosion due to oxygen – leading to formation of oxides

ii] corrosion due to other gases – leading to formation of other compounds

iii] corrosion due to other corrosive liquids – such as other molten metals.

## 1. corrosion due to oxygen :-

- This type of corrosion occurs due to attack of atmospheric oxygen on metal surface either at low or high temprerature. forming metal oxide as

$$\text{Metal} + \text{oxygen} \longrightarrow \text{Metal oxide}$$

The nature and stability of oxide film depends on temprecture, humidity and athmosphere conditions
- The corrosion depends upon the type of the oxide film.
- It can be classified as.
  i] stable oxide film — a] porous b] non-porous
  ii] unstable oxide film
  iii] volatile oxide film.

## Pilling - Bedworth rule

According to it
- an oxide is protective or non-porous, if the volume of the oxide is at least as great as the volume of the metal form which is formed
- on the other hand, if the volume of oxide is less than the volume of the metal the oxide layer is porous and hence non-protective. because it cannot prevent the access of oxygen to the fresh metal surface below

## 1] Stable oxide film

- If the oxide film is stable, it acts as a protective coating to the metal surface.
- A stable layer of metal oxide is formed which is impervious and this protects the metal from the furthur corrosion.

### (a) Porous oxide layer

- If oxide film formed on metal surface is porous in nature, the rate of further corrosion is not reduced much.
- e.g. alkali metal such as Li, Na, K or alkaline earth metals such as Ca, Mg etc.

- These metals react with oxygen forming oxi as $Na_2O$, $K_2O$, $CaO$, $MgO$ etc.
- where volume of oxide formed is less than the of the reacted metal.
- Hence porous oxide scale permits free access of oxyge to the underlying metal surface. (through cracks and pores) thus corrosion continues not stop.

(b) Nonporous oxide film

- If the oxide film formed is non-porous in nature the rate of corrosion reduces considerably and in most of cases corrosion of metal stops.
- The metals like Al, Cr, Cu, Pb, Sn etc forms their oxides such as $Al_2O_3$, $Cr_2O_3$ etc.
- where volume of oxide formed is greater than the volume of the reacted metal
- Hence in non-porous oxide film not leaving any access for further attack of oxygen thereby stopping further corrosion.

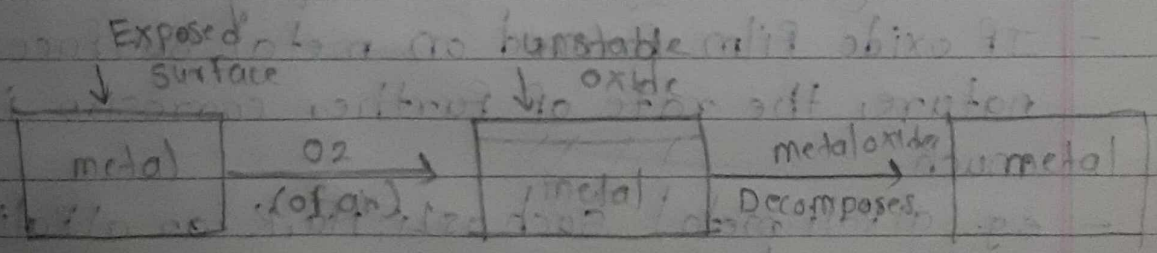
$$2Zn + O_2 \rightarrow 2ZnO$$
$$2Al + 3/2\, O_2 \rightarrow Al_2O_3$$

(2) Unstable oxide film :-

- If the oxide film formed is unstable it decomposes on the metal surface forming back the metal and oxygen.

Metal + oxygen ⟶ metal oxide ⟶ metal + oxygen

- Thus oxidation corrosion does not occur
- e.g. gold, platinum, silver. $4Ag + O_2 \rightarrow 2Ag_2O$

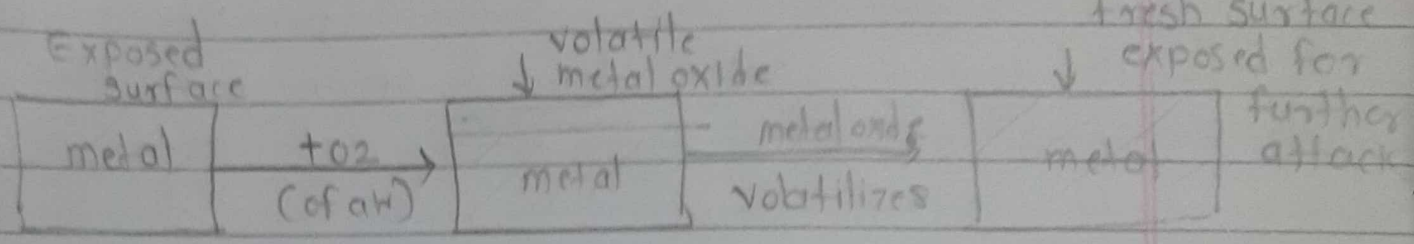



Unstable oxide layer

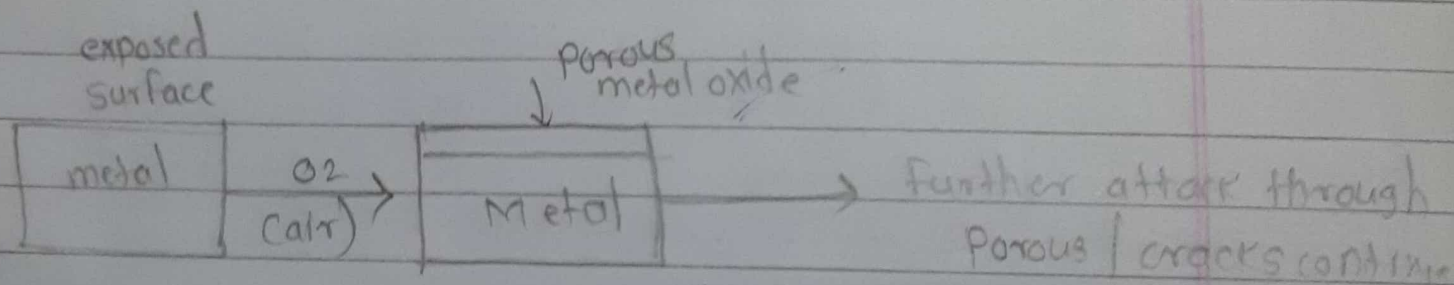
## Volatile oxide film

If the metal oxide film is volatile the oxide is volatalize as soon as it is formed and hence the underlying metal surface is exposed for further attack

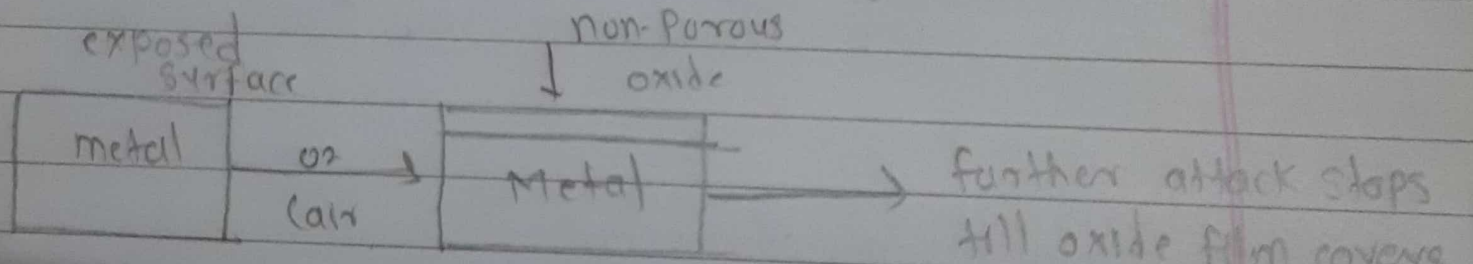
eg. Molybdenum oxide ($MoO_3$) film is volatile




Volatile oxide layer




Porous oxide layer




Non-Porous oxide layer

## Wet corrosion or electrochemical corrosion

- when metal surface is an immediate contact of aqueous acidic/ alkaline/ neutral electrolyte solutions. the short circuited galvanic cells get set all along the surface of the metal
- Wet corrosion also known as immersed corrosion occurs under the following conditions.-
  1. when a conducting liquid is in contact with the metal
  2. when two different metals or alloys are partially or completely immersed in a solution.

- According to electrochemical theory, This type of corrosion occurs where a conducting liquid is in contact with the metal. This corrosion occurs due to the existence of seprate anodic and cathodic parts between which current flows through conducting solution
- At Anodic area oxidation reaction occurs there by destroying the anodic metal either by dissolution or formation of compounds. Hence corossion always occurs at anodic parts.

### Mechanism

- Electrochemical corrosion involves flow of electron between Anode and cathode
- The anodic reaction involves dissolution of metal liberating free electrons.

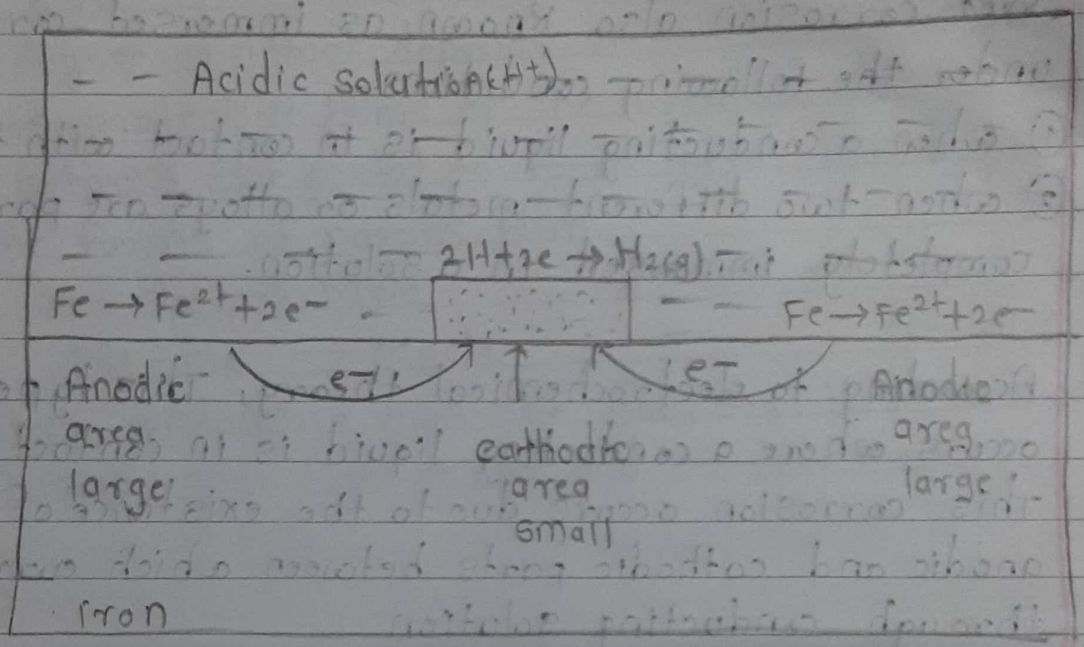
$$M \longrightarrow M^{n+} + ne^-$$

The cathodic reaction consumes electrons with either
- evolution of hydroden or
- absorption of oxygen

which depends on nature of corrosive enviroment

## ① Hydrogen evolution mechanism:-

This type of corrosion occurs in acidic medium e.g. considering metal Fe, anodic reaction is dissolution of ions as ferrous ions with liberation of electrons.




Anode $\longrightarrow$ $Fe \longrightarrow Fe^{2+} + 2e^-$ (oxidation)

- The electron released flow through the metal from anode to cathode, where $H^+$ ion of acidic solution are eliminated as hydrogen gas.

- cathode $\rightarrow$ $2H^+ + 2e^- \longrightarrow H_2$ (Reduction)

The overall reaction is $Fe + 2H^+ \longrightarrow Fe^{2+} + H_2$.

- This type of corrosion, metal ion dissolve in solution and hydrogen ions come out of solution in the form of gas
- All metals above hydrogen in electrochemical series have a tendency to get dissolved in acidic solution with simultaneous evolution of $H_2$ gas
- The anodes are large areas, whereas cathodes are small areas.

## Absorption of oxygen :-

- For example, rushting of iron in neutral aqueous solution of electrolyte in presence of atmospheric oxygen
- Usually surface of iron is coated with a thin film of iron oxide.
- If the film develops cracks, anodic area are created on the surface, while metal parts act as cathode
- It shows that anode are small areas while the rest metallic part form large area. cathodes.
- The electron flow from anode to cathode through iron metal.

At Anode → $Fe \rightarrow Fe^{2+} + 2e^-$ (oxidation)

At cathode $\frac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$ (Reduction)

~~overall~~

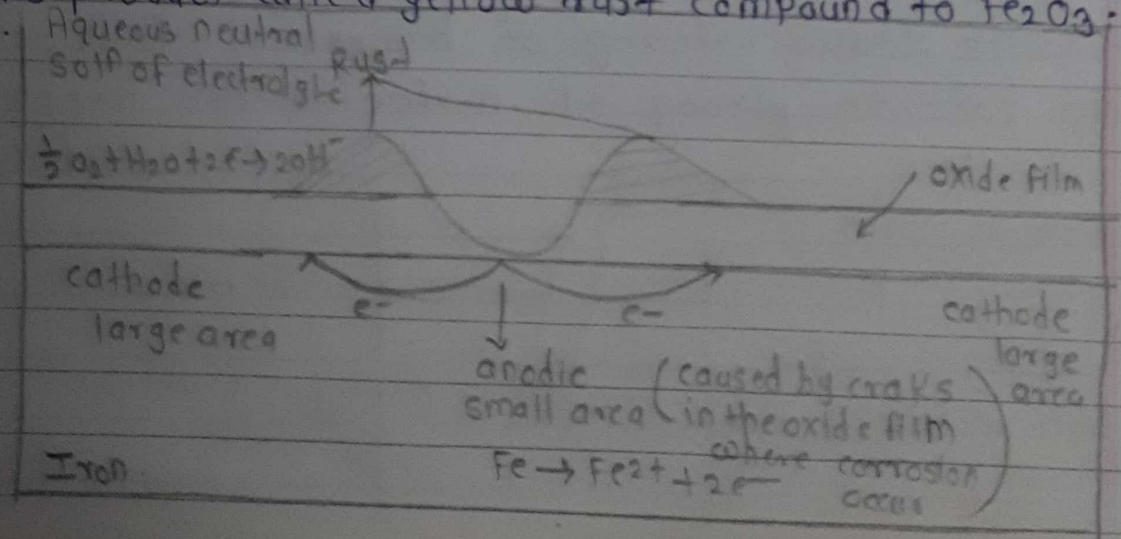
The ferrous ions at anode and hydroxyl ions at the cathode diffuse towards each other i.e. $Fe^{2+}$ ions diffuse towards eanode and $OH^-$ ions diffuse towards cathode and then they combine to form ferrous hydroxide

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2 \downarrow$$

(brown rust)

If oxygen in excess. ferrous hydroxide is easily oxidized to ferric hydroxide.

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \downarrow$$

(yellow rust)

This product called yellow rust compound to $Fe_2O_3 \cdot 3H_2O$.